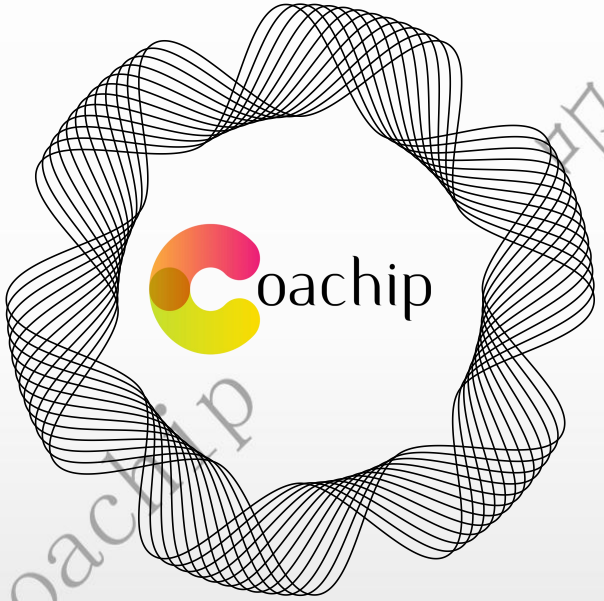


SV的数组约束

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课程内容及目标

课程内容：

- 约束数组的size()函数
- 约束数组的sum()函数
- 数组约束的错误例子分析



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约束数组的size()函数

- 用来统计动态数组或者队列的元素个数

```
class dyn_size;
    rand logic [31:0] d[];
    constraint d_size {d.size() inside {[1:10]};}
endclass
```

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约束数组的sum()函数

- 用来计算动态数组或者队列的所有元素的和

```
class dyn_sum;
    rand logic [7:0] d[];
    constraint d_sum {d.sum() ==9'd100;}
endclass
```

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System Verilog数组约束

- Bad Example1

requirement：

1~8 unsigned elements;

each one is inside [0:255];

sum is less than 1024.

```
sum= -53, val= -58 -85-115 112-101 -62
```

```
class bad_sum1;
    rand byte len[];
    constraint c_len
    {
      len.sum < 1024;
      len.size() inside {[1:8]};
    }

    function void display();
      $write("sum=%4d, val=", len.sum);
      foreach(len[i]) $write("%4d ", len[i]);
    endfunction
endclass
```

产生了一些lengths 但是和有时候是负数或者总是小于1024

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System Verilog数组约束

- Bad Example2

requirement：

1~8 unsigned elements;

each one is inside [0:255];

sum is less than 1024.

sum= 1530, val=255 255 255 255 255 255

```
class bad_sum2;
  rand bit [7:0] len[];
  constraint c_len
  {
    len.sum < 1024;
    len.size() inside {[1:8]};
  }

  function void display();
    $write("sum=%4d", len.sum);
    foreach(len[i]) $write("%4d", len[i]);
  endfunction
endclass
```

结果仍然使用8bit, 可能会溢出

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System Verilog数组约束

- Bad Example3

requirement：

1~8 unsigned elements;

each one is inside [0:255];

sum is less than 1024.

```
sum= 245, val=1348956995 3748256598 985546882 2507174362
```

```
class bad_sum3;
  rand int len[];
  constraint c_len
  {
    len.sum < 1024;
    len.size() inside {[1:8]};
  }

  function void display();
    $write("sum=%4d", len.sum);
    foreach(len[i]) $write("%4d", len[i]);
  endfunction
endclass
```

两个很大的数值, 相加后可能会溢出,从而变成一个小于1024的数

System Verilog数组约束

- Good Example

requirement：

1~8 unsigned elements;

each one is inside [0:255];

sum is less than 1024.

```
sum= 976, val= 233 187 44 157 201 81 73
```

```
class good_sum;
  rand int len[];
  constraint c_len
  {
    foreach(len[i]) len[i] inside {[1: 255]};
    len.sum < 1024;
    len.size() inside {[1:8]};
  }

  function void display();
    $write("sum=%4d", len.sum);
    foreach(len[i]) $write("%4d", len[i]);
  endfunction
endclass
```

len array 可以大于10bit或者更多, 但必须是无符号整形

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练习

requirement：

1~8 unsigned elements;

each one is inside [0:255];

sum is less than 1024.

检查随机值是否在合理范围之内。

测试前面的bad example

```
class good_sum;
  rand int len[];
  constraint c_len
  {
    foreach(len[i]) len[i] inside {1: 255};
    len.sum < 1024;
    len.size() inside {[1:8]};
  }

  function void display();
    $write("sum=%4d", len.sum);
    foreach(len[i]) $write("%4d", len[i]);
  endfunction
endclass
```

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Q&A

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